

Name:Prachi Santosh Jadhav

Kubernetes_Basics_Assignment

Q1

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ minikube start
W0908 21:27:11.483361 37207 main.go:294] unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/prachi/.docker/context/meta/37a8eec1ce19687d132fe29851dca629d164e2c4958ba141d3f4133833f0680f/meta.json: no such file or directory
W0908 21:27:11.483441 37207 main.go:295] Try running 'docker context use default' to resolve the above error
🐳 minikube v1.38.1 on Ubuntu 26.04
🔑 Automatically selected the docker driver
👉 Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
👉 Using Docker driver with root privileges
🔑 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.50 ...
🔑 Creating docker container (CPUs=2, Memory=3072MB) ...
👉 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔑 Configuring bridge CNI (Container Networking Interface) ...
🔑 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Enabled addons: storage-provisioner, default-storageclass
👉 Done! kubectll is now configured to use "minikube" cluster and "default" namespace by default
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll get nodes
NAME                STATUS    ROLES    AGE   VERSION
minikube            Ready     control-plane  33s   v1.35.1
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll version --client
Client Version: v1.36.3
Kustomize Version: v5.8.1
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
```

Q2

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll run myweb --image=nginx
pod/myweb created
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll get pods
NAME    READY   STATUS    RESTARTS   AGE
myweb   0/1     ContainerCreating    0          7s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll get pods
NAME    READY   STATUS    RESTARTS   AGE
myweb   1/1     Running   0          49s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectll describe pod myweb
Name:         myweb
Namespace:    default
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Tue, 08 Sep 2026 21:29:29 +0530
Labels:       run=myweb
Annotations:   <none>
Status:       Running
IP:           10.244.0.3
IPs:          10.244.0.3
Containers:
  myweb:
    Container ID:  docker://863a36afb7d3abd8d5052f7acf11015eff0a32a7042e3594ffe35ba75706b
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:05b8cb0c354a4ab024ea67dc69b46d50762cde728a347a5b656e0fb3d7c4
    Port:         <none>
    Host Port:    <none>
    State:        Running
      Started:    Tue, 08 Sep 2026 21:29:49 +0530
    Ready:        True
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-n6s87 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers    True
  Initialized         True
  Ready               True
  ContainersReady     True
  PodScheduled        True
Volumes:
  kube-api-access-n6s87:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    Optional:            false
    DownwardAPI:        true
QoS Class:             BestEffort
Node-Selectors:         <none>
Tolerations:            node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                        node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
```

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx: ~
Initialized: True
Ready: True
ContainersReady: True
PodScheduled: True
Volumes:
  kube-apl-access-n6s87:
    Type: Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations:
  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type Reason Age From Message
  ---
  Normal Scheduled 67s default-scheduler Successfully assigned default/myweb to minikube
  Normal Pulling 66s kubelet spec.containers(myweb): Pulling image "nginx"
  Normal Pulled 47s kubelet spec.containers(myweb): Successfully pulled image "nginx" in 19.301s (19.301s including waiting). Image size: 170059427 bytes.
  Normal Created 47s kubelet spec.containers(myweb): Container created
  Normal Started 46s kubelet spec.containers(myweb): Container started
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl logs myweb
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/08 15:59:50 [notice] 1#1: using the "epoll" event method
2026/09/08 15:59:50 [notice] 1#1: nginx/1.31.5
2026/09/08 15:59:50 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/08 15:59:50 [notice] 1#1: OS: Linux 7.0.0-30-generic
2026/09/08 15:59:50 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/09/08 15:59:50 [notice] 1#1: start worker processes
2026/09/08 15:59:50 [notice] 1#1: start worker process 29
2026/09/08 15:59:50 [notice] 1#1: start worker process 30
2026/09/08 15:59:50 [notice] 1#1: start worker process 31
2026/09/08 15:59:50 [notice] 1#1: start worker process 32
2026/09/08 15:59:50 [notice] 1#1: start worker process 33
2026/09/08 15:59:50 [notice] 1#1: start worker process 34
2026/09/08 15:59:50 [notice] 1#1: start worker process 35
2026/09/08 15:59:50 [notice] 1#1: start worker process 36
2026/09/08 15:59:50 [notice] 1#1: start worker process 37
2026/09/08 15:59:50 [notice] 1#1: start worker process 38
2026/09/08 15:59:50 [notice] 1#1: start worker process 39
2026/09/08 15:59:50 [notice] 1#1: start worker process 40
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
```

Q3

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx: ~
Volumes:
  kube-apl-access-n6s87:
    Type: Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations:
  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type Reason Age From Message
  ---
  Normal Scheduled 67s default-scheduler Successfully assigned default/myweb to minikube
  Normal Pulling 66s kubelet spec.containers(myweb): Pulling image "nginx"
  Normal Pulled 47s kubelet spec.containers(myweb): Successfully pulled image "nginx" in 19.301s (19.301s including waiting). Image size: 170059427 bytes.
  Normal Created 47s kubelet spec.containers(myweb): Container created
  Normal Started 46s kubelet spec.containers(myweb): Container started
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl logs myweb
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/08 15:59:50 [notice] 1#1: using the "epoll" event method
2026/09/08 15:59:50 [notice] 1#1: nginx/1.31.5
2026/09/08 15:59:50 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/08 15:59:50 [notice] 1#1: OS: Linux 7.0.0-30-generic
2026/09/08 15:59:50 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/09/08 15:59:50 [notice] 1#1: start worker processes
2026/09/08 15:59:50 [notice] 1#1: start worker process 29
2026/09/08 15:59:50 [notice] 1#1: start worker process 30
2026/09/08 15:59:50 [notice] 1#1: start worker process 31
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2026/09/08 15:59:50 [notice] 1#1: start worker process 34
2026/09/08 15:59:50 [notice] 1#1: start worker process 35
2026/09/08 15:59:50 [notice] 1#1: start worker process 36
2026/09/08 15:59:50 [notice] 1#1: start worker process 37
2026/09/08 15:59:50 [notice] 1#1: start worker process 38
2026/09/08 15:59:50 [notice] 1#1: start worker process 39
2026/09/08 15:59:50 [notice] 1#1: start worker process 40
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl delete pod myweb
pod "myweb" deleted from default namespace
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
No resources found in default namespace.
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
```

Q4

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/08 15:59:50 [notice] 1#1: using the "epoll" event method
2026/09/08 15:59:50 [notice] 1#1: nginx/1.31.5
2026/09/08 15:59:50 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/08 15:59:50 [notice] 1#1: OS: Linux 7.0.0-30-generic
2026/09/08 15:59:50 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/09/08 15:59:50 [notice] 1#1: start worker processes
2026/09/08 15:59:50 [notice] 1#1: start worker process 29
2026/09/08 15:59:50 [notice] 1#1: start worker process 30
2026/09/08 15:59:50 [notice] 1#1: start worker process 31
2026/09/08 15:59:50 [notice] 1#1: start worker process 32
2026/09/08 15:59:50 [notice] 1#1: start worker process 33
2026/09/08 15:59:50 [notice] 1#1: start worker process 34
2026/09/08 15:59:50 [notice] 1#1: start worker process 35
2026/09/08 15:59:50 [notice] 1#1: start worker process 36
2026/09/08 15:59:50 [notice] 1#1: start worker process 37
2026/09/08 15:59:50 [notice] 1#1: start worker process 38
2026/09/08 15:59:50 [notice] 1#1: start worker process 39
2026/09/08 15:59:50 [notice] 1#1: start worker process 40
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl delete pod myweb
pod "myweb" deleted from default namespace
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
No resources found in default namespace.
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl create deployment web --image=nginx --replicas=2
deployment.apps/web created
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-zhj9p 1/1 Running 0 6s
web-68d995574f-8qspk 1/1 Running 0 6s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl delete pod web-68d995574f-zhj9p
pod "web-68d995574f-zhj9p" deleted from default namespace
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-8qspk 1/1 Running 0 61s
web-68d995574f-9dxds 1/1 Running 0 16s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl scale deployment web --replicas=4
deployment.apps/web scaled
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-2ls18 1/1 Running 0 6s
web-68d995574f-8qspk 1/1 Running 0 81s
web-68d995574f-9dxds 1/1 Running 0 36s
web-68d995574f-9k1lj 1/1 Running 0 6s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
web 4/4 4 4 90s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
```

Q5

```
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
2026/09/08 15:59:50 [notice] 1#1: start worker process 34
2026/09/08 15:59:50 [notice] 1#1: start worker process 35
2026/09/08 15:59:50 [notice] 1#1: start worker process 36
2026/09/08 15:59:50 [notice] 1#1: start worker process 37
2026/09/08 15:59:50 [notice] 1#1: start worker process 38
2026/09/08 15:59:50 [notice] 1#1: start worker process 39
2026/09/08 15:59:50 [notice] 1#1: start worker process 40
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl delete pod myweb
pod "myweb" deleted from default namespace
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
No resources found in default namespace.
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl create deployment web --image=nginx --replicas=2
deployment.apps/web created
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-zhj9p 1/1 Running 0 6s
web-68d995574f-8qspk 1/1 Running 0 6s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl delete pod web-68d995574f-zhj9p
pod "web-68d995574f-zhj9p" deleted from default namespace
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-8qspk 1/1 Running 0 61s
web-68d995574f-9dxds 1/1 Running 0 16s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl scale deployment web --replicas=4
deployment.apps/web scaled
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
web-68d995574f-2ls18 1/1 Running 0 6s
web-68d995574f-8qspk 1/1 Running 0 81s
web-68d995574f-9dxds 1/1 Running 0 36s
web-68d995574f-9k1lj 1/1 Running 0 6s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
web 4/4 4 4 90s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl expose deployment web --port=80 --type=NodePort
service/web exposed
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 9h36s
web NodePort 10.101.172.122 <none> 80:32576/TCP 16s
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ minikube service web
W0908 21:37:41.898175 47670 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/prachi/.docker/contexts/meta/37a8ee1ce19687d132Fe29851dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0908 21:37:41.898223 47670 main.go:295] Try running 'docker context use default' to resolve the above error

NAMESPACE NAME TARGET PORT URL
default web 80 http://192.168.49.2:32576

$ Opening service default/web in default browser...
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$ Gtk-Message: 21:37:43.746: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
prachi@prachi-Victus-by-HP-Gaming-Laptop-15-fb0xxx:~$
```

